

Stable Causal Targeting with Machine Learning

Application to a Behavioral Field Experiment in Consumer Finance

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Motivation: experiments estimate averages, but responses vary

- Firms run randomized field experiments to estimate **average treatment effects** (ATE).
- But responses are often **heterogeneous**, and in behavioral settings also **noisy**.
- Flexible ML estimators of CATEs can **overfit** training folds, show **weak calibration** on new data, and yield **unstable** prioritization rules.
- Aggressive subgroup search without adjustment finds effects that do **not** replicate.

Goal

A practical, stability-first workflow for subgroup targeting that survives repeated data perturbations and an untouched holdout.

Background: StaDISC and the PCS framework

StaDISC — *Stable Discovery of Interpretable Subgroups via Calibration* (Dwivedi et al., 2020)

- Originally built for subgroup discovery in **clinical trials**.
- True CATE unobserved $\Rightarrow$ cannot use AUC; use **calibration** instead.
- Bin units by CATE estimates; compare bin-mean CATE vs. difference-in-means.

PCS framework — *Predictability, Computability, Stability* (Yu & Kumbier, 2018, 2020):

Predictability = out-of-sample accuracy; **Computability** = feasible & reproducible;
Stability = robust to perturbations in data, model, and sampling.

This paper

Extend StaDISC from clinical trials to a **consumer-banking** field experiment with smaller effects and noisier outcomes.

What we add to StaDISC

- **Setting:** behavioral finance (low signal-to-noise) instead of clinical trials.
- **Feature selection:** Lasso pre-screening (519 $\rightarrow$ 14 features) — not in original StaDISC.
- **Selection rule:** replace the t -statistic grid with a **calibration-slope nested-ensemble path**:
 1. order candidate estimators by validation-fold CalSlope,
 2. build nested equal-weight mean ensembles along that order,
 3. pick the ensemble maximizing the top-10% targeting contrast $\overline{\Delta}_{0.10}$.
- **Validation:** evaluate the selected ensemble on a completely untouched holdout.

The YK overdraft experiment

Setting: unshrouding overdraft costs in Turkey

Alan, Cemalcular, Karlan & Zinman (2018) — Yapi Kredi (YK), one of Turkey's largest banks. 108,000 marginal overdrafters randomized to text-message arms; outcome = overdraft usage.

- **Availability reminder** $\Rightarrow$ *increased* usage (makes overdraft salient).
- **Discount message** (50% off 60% APR) $\Rightarrow$ *decreased* usage (unshrouds the cost).
- **Bundled messages** (discount + bill-payment / debit-card) $\Rightarrow$ *further* decrease.

Shrouded-attributes logic (Gabaix & Laibson, 2006): unshrouding may produce **heterogeneous** responses —

- *financially sophisticated* consumers already aware of overdraft,
- *liquidity-constrained* consumers with acute short-term needs.

Data and sample design

Main comparison: bill-payment bundled message vs. baseline availability reminder.

- Full sample: **36,024** customers
 - 17,981 treated (bundled)
 - 18,043 control (reminder)
- Outcome: binary overdraft usage.
- Full-sample ATE = -0.0138 (SE 0.0049).

Split (stratified by treatment):

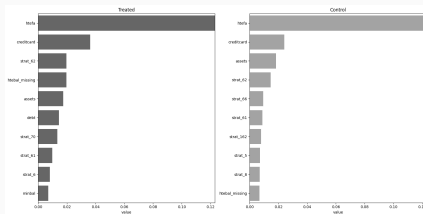
- **20% holdout** — 7,205 customers; touched only at the very end.
- **80% train/val** — 28,819 customers.

Cross-validation: three stratified 4-fold splits

$(cv_{\text{orig}}, cv_0, cv_1) \Rightarrow$ 12 training-validation fold pairs.

Feature engineering & selection

- Covariates: strata, reminder frequency, campaign type, prior activations, assets, deposits, debt, balances, credit-card usage, ...
- Missing values $\Rightarrow$ missingness dummies + zero imputation $\Rightarrow$ **519 features**.
- **Lasso selection** (treated / control separately, standardized): top-10 by $|\text{coef}|$ each $\Rightarrow$ union = **14 features** — extends StaDISC.



CATE models & calibration check

23 CATE estimation models

Family	Idea	#
S-learner	One model on (X, Z) ; $\text{CATE} = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$	2
T-learner	Separate $\hat{\mu}_1, \hat{\mu}_0$; $\text{CATE} = \hat{\mu}_1 - \hat{\mu}_0$	4
X-learner	Impute ITEs, then regress (Künzel et al., 2019)	4
R-learner	Residualize Y, Z on X , then weighted regression (Nie & Wager, 2021)	9
Causal tree / forest	Honest partitions / ensembles (Athey & Imbens; Wager & Athey)	4
Total		23

Base learners: **Lasso**, penalized logistic, RF, XGB.

Hyperparameters tuned by 4-fold CV on cv_{orig} (200 random configs each); fixed on cv_0, cv_1 .

Calibration-based reality check ($\mathcal{R}_C^2$)

True CATE $\tau(x_i)$ is unobserved $\Rightarrow$ AUC is not available. Use **calibration**.

Key identity for any subgroup G of the feature space:

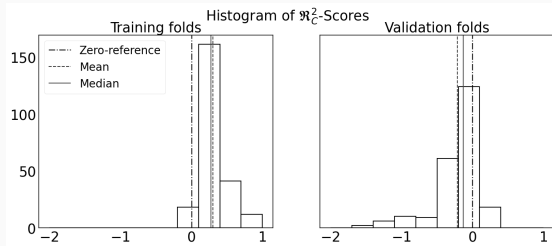
$$\mathbb{E}[Y(1) - Y(0) \mid X \in G] = \mathbb{E}[\tau(X) \mid X \in G].$$

- Bin units by CATE estimates into 10 deciles $\mathcal{G}_1, \dots, \mathcal{G}_{10}$.
- In each bin compare **mean model CATE** $\bar{\mathfrak{m}}$ vs. **difference-in-means** $\hat{\tau}$.

$$\text{Cal-Score} = \sum_j \frac{|\mathcal{G}_j \cap \mathcal{S}|}{|\mathcal{S}|} |\bar{\mathfrak{m}}_{\mathcal{G}_j \cap \mathcal{S}} - \hat{\tau}_{\mathcal{G}_j \cap \mathcal{S}}|, \quad \mathcal{R}_C^2 = 1 - \frac{\text{Cal-Score}(\bar{\mathfrak{m}})}{\text{Cal-Score}(\hat{\tau}_{\text{ATE}})}.$$

$\mathcal{R}_C^2 \rightarrow 1$ good calibration; $\mathcal{R}_C^2 \leq 0$ no better than a constant ATE $\Rightarrow$ overfitting.

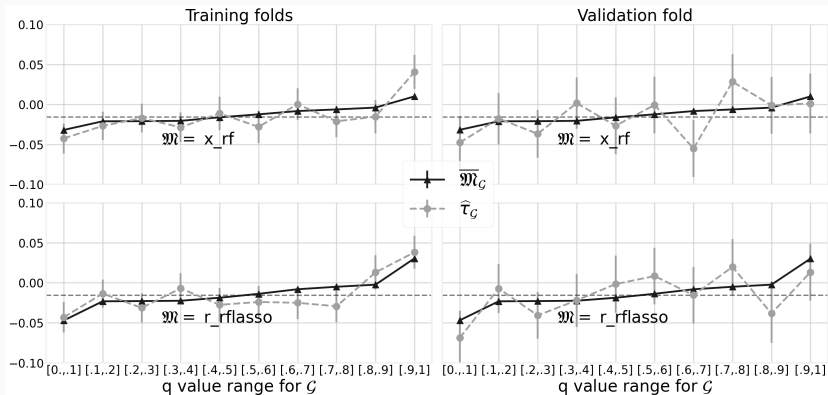
Result: poor global calibration = overfitting



- Training $\mathcal{R}_C^2$: mean 0.30, median 0.28.
- Validation $\mathcal{R}_C^2$: mean -0.21 , median -0.13 .
- Systematic gap = models fit noise, not real heterogeneity.
- Echoes the original StaDISC finding on clinical data.

Bottom line: *globally*, all 23 CATE models fail to calibrate on held-out folds.

Calibration plots: reasonable in-sample, breaks out-of-sample



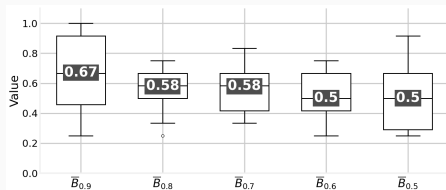
x_rf (X-learner+RF) and $r_rflasso$ (R-learner, RF→Lasso). Triangles = model CATE; circles = difference-in-means ± 1 SE. Left: training; Right: validation. Dashed = overall ATE (-0.0158).

Quantile-based top subgroups: look for *local* signal

Global calibration fails, but a model may still **rank** the top units correctly.

For quantile $q \in \{0.9, 0.8, 0.7, 0.6, 0.5\}$ define the **top subgroup** $\tilde{\mathcal{G}}_q^c$ and the indicator

$$B_q = \mathbf{1}\left\{\hat{\tau}_{\tilde{\mathcal{G}}_q \cap \mathcal{S}_{\text{VF}}} \leq \hat{\tau}_{\tilde{\mathcal{G}}_q^c \cap \mathcal{S}_{\text{VF}}}\right\}, \quad \bar{B}_q = \frac{1}{12} \sum_{f=1}^{12} B_q.$$



Most models are poorly calibrated *and* unstable; $q = 0.9$ is the best-behaved $\Rightarrow$ focus on the **top 10%**.

Selection & validation

Selection rule: a calibration-slope nested-ensemble path

Step 1 — Order by CalSlope. Cut training-fold CATE into 5 strata; regress observed stratum ATE $\hat{\theta}_{mfk}$ on mean predicted CATE $\bar{\mu}_{mfk}$, take the OLS slope $\text{CalSlope}_{mf} = \hat{\beta}_{mf}$, average over folds: $\text{CalSlope}_m = \frac{1}{|\mathcal{F}_m|} \sum_f \text{CalSlope}_{mf}$. Order estimators from largest to smallest CalSlope_m .

Step 2 — Nested mean ensembles. Ensemble 1 = top estimator; ensemble 2 = top 2; ...

Step 3 — Pick by top-10% contrast. For ensemble e , fold f :

$$\Delta_{ef,0.10} = \hat{\tau}(\mathcal{G}_{ef}^{0.10}) - \hat{\tau}(\mathcal{R}_{ef}^{0.10}), \quad \bar{\Delta}_{0.10}(e) = \frac{1}{|\mathcal{F}_e|} \sum_f \Delta_{ef,0.10}.$$

Select the ensemble with the **largest** $\bar{\Delta}_{0.10}$ (ties $\Rightarrow$ fewer members). Public rule: CalSlope to order, $\bar{\Delta}_{0.10}$ to select.

Selected ensemble on the YK data

	Value
Ordering statistic	CalSlope
Candidate path	nested ensembles
Selection statistic	$\bar{\Delta}_{0.10}$
Selected (2 of 23)	
x_rf	✓
r_rflasso	✓
Targeting quantile	top 10%

- Despite negative full-sample ATE, the top-10% subgroup shows a **positive** effect.
- Two very different families (X-learner, R-learner) agree $\Rightarrow$ stability signal.
- The ensemble is small (2 members), reducing overfitting risk vs. larger blends.

Final validation on the untouched holdout

	Train-Val	Holdout
Total sample size	28,819	7,205
Top-10% subgroup size	2,882	723
Subgroup ATE (SE)	0.0407 (0.0175)	0.0617 (0.0346)
t-statistic	2.32	1.78
95% CI	[0.0064, 0.0750]	[−0.0061, 0.1296]
90% CI	[0.0119, 0.0695]	[0.0048, 0.1187]

- Holdout top-10% ATE = +6.17 pp, **significant at the 10% one-sided level**.
- Effect *grows* from train-val to holdout $\Rightarrow$ not an in-sample artifact.
- Full-sample ATE was −1.38 pp: targeting flips the sign for the right subgroup.

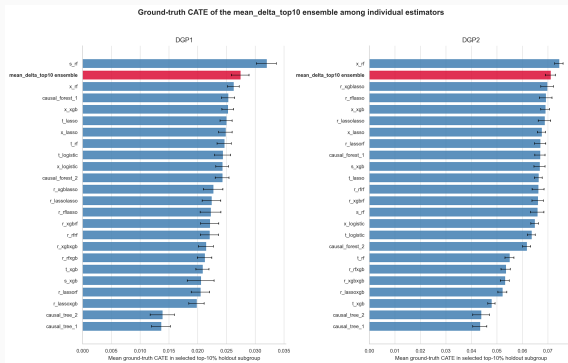
The other two arms do **not** replicate

Same procedure applied to the other treatment-arm comparisons:

Treatment arm	Selected ensemble	Holdout ATE (SE), t
Debit-card bundled msg	s_rf x_rf x_lasso	-0.0724 (0.0362), -2.00
Availability reminder vs. none	t_lasso x_logistic	-0.0119 (0.0306), -0.39

- Neither produces a positive holdout effect.
- The **bill-payment bundled message** is the only arm with a positive top-10% holdout signal.
- Stability screening prevents overconfident targeting recommendations.

Simulation: $\bar{\Delta}_{0.10}$ picks a near-best ensemble



Ground-truth top-10% CATE of the ensemble selected by $\bar{\Delta}_{0.10}$ vs. all 23 individual estimators, across two simulation DGPs. The selected ensemble attains true-CATE rank ≤ 2 in both designs.

Conclusion

Takeaways

- In low-signal behavioral experiments, **global CATE estimation fails**: all 23 models are poorly calibrated out-of-sample.
- Yet a **calibration-slope nested-ensemble** rule identifies a top-10% subgroup with a **positive holdout effect** (+6.17 pp, $p < 0.10$ one-sided).
- The rule is public and simple: CalSlope to order, $\bar{\Delta}_{0.10}$ to select.
- Other arms do not replicate $\Rightarrow$ stability screening prevents overconfident targeting.

PCS as a template

Smaller effects and noisier outcomes than the original clinical-trial StaDISC — yet calibration reality checks, validation-fold selection, and a quantile-based top subgroup still deliver a useful targeting rule.

Thank you

Questions & discussion welcome